

Ex) If $f(x) = x^4 + 12x^3 - 2x^2 - 420x$, find the intervals of increase or decrease. Hint: $f'(3) = 0$ (use long division)

$$f'(x) = 4x^3 + 36x^2 - 4x - 420$$

$$x-3 \overline{4x^3 + 36x^2 - 4x - 420}$$

$$-(4x^3 - 12x^2)$$

$$48x^2 - 4x - 420$$

$$-(48x^2 - 144x)$$

$$140x - 420$$

$$-(140x - 420)$$

OR

$$4x^2 + 48x + 140$$

$$= 4(x^2 + 12x + 35)$$

$$= 4(x+7)(x+5)$$

$$\therefore 4(x-3)(x+7)(x+5) = 0$$

$\Rightarrow$ critical #'s are

$$3, -7, -5$$

$\Rightarrow$ 4 intervals

Interval	$(-\infty, -7)$	$(-7, -5)$	$(-5, 3)$	$(3, \infty)$
$x-3$	-	-	-	+
$x+7$	-	+	+	+
$x+5$	-	-	+	+
$f'(x)$	-	+	-	+
$f(x)$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$
	D	I	D	I
Increasing		$(-7, -5) \cup (3, \infty)$		
Decreasing	$(-\infty, -7) \cup (-5, 3)$			

Ex) If $f(x) = -3x^{\frac{5}{3}} + 15x^{\frac{2}{3}}$ find the intervals of increase or decrease.

$$f'(x) = -5x^{\frac{2}{3}} + 10x^{-\frac{1}{3}}$$

$$= -5x^{-\frac{1}{3}}(x-2)$$

$\therefore$ critical #'s at

$$x=2 \text{ and } x=0$$

$\uparrow$

$$f'(2) = 0$$

$\uparrow$

$$f'(0) \text{ DNE}$$

Interval	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
-5	-	-	-
$x^{\frac{1}{3}}$	-	+	+
$x-2$	-	-	+
$f'(x)$	-	+	-
$f(x)$	$\downarrow$	$\uparrow$	$\downarrow$
	D	I	D

Increasing $(0, 2)$
Decreasing $(-\infty, 0) \cup (2, \infty)$